

Maximal Workflow - Kolbe electrolysis - Flow



Category Experimental Procedures

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Labels: [Kolbe electrolysis](#)

General Remarks

In the maximal workflow: (at least) three replicates are performed independently !

Replicates might vary regarding:

- in-situ pH/conductivity measurements (& therefore: 1- or 3-neck reactor)
- online gasphase measurements

Preparation of solution(s)

- Prepare a anolyte reservoir (100 mL Flask) with stirrer and label
- **Weigh the empty flask**
- Add 50% LM to the flask
- Add the carboxylic acid(s)
- Adjust the pH to planned pH (if you use a base as supporting electrolyte (e.g. Na_2CO_3): adjust to pH = 7 (via instruction below), add the base, and adjust to pH(via instruction below))
 - NaOH pellets (rinsed with distilled water), 10 M NaOH, 0.1 M NaOH & 3 M H_2SO_4 used to adjust the pH (pH measurement during adjustment)
 - Can be calculated / values from similar setups can be used as base
- Add the remaining LM, if necessary
- Add the supporting electrolyte, if necessary (solid & neutral, e.g. Na_2SO_4)
- Measure conductivity & pH
- Take 1 mL of sample from solution as t0 (into weighed empty HPLC vial) and weigh the sample
- **Weigh the full flask**

For a two-chamber-set-up // Catholyte preparation of 0.25 M Na_2SO_4 :

- Prepare a stock solution (1 L Flask) with stirrer and label
- Add 50% LM (MilliQ Water)
- Add the electrolyte (solid Na_2SO_4) to get a concentration of 0.25 M Na_2SO_4
- Add the remaining LM

- Measure pH & conductivity

Set-Up for electrolysis

- Flush the gas bag 2-3 times with nitrogen (fill with nitrogen and empty using a water jet pump) and connect to the mass flow meter (MFM; keep the connection MFM - gas bag closed)

For online gasphase measurements:

- BakeOut the microGC for 20 min
- Load latest calibrated Method: "MicroGC_standard_method_calibrated_0726" (or latest version)

For a two-chamber-set-up:

- Fill 200 mL of the catholyte in the catholyte reservoir with stirrer and label
- Screw the flow reactor with 2 Pt electrodes, membrane and gaskets tightly together
- Mount it upright, that the flow reactor outlet is above the inlet and the long side is parallel to the floor
- Test the flow reactor for leaks by running the system with MilliQ water for 1-2 minutes, if no water comes out of the flow reactor, flush the system with nitrogen
- Place and secure the reaction solution(s) in the anolyte and catholyte reservoir on the stirring plate
- Connect the pH meter and conductivity sensor + temperature to the anolyte reservoir
- Set the stirring speed to 800 RPM and set the correct flow rate
- Connect the electrodes (WE = +/CE = -) to the DC source and set to the desired conditions (150 mA cm⁻²)
- Connect the flow reactor inlets and outlets with the solution reservoirs and the pump
- Turn on the pump with the desired settings and let the flow reactor fill up
- Let the system run for 1-2 min and check for leaks
- Connect the gas inlet to the nitrogen line and open the gas inlet
- Flush the solution with nitrogen for 10 minutes (moderate bubbling) and then close the inlet
- After flushing: Connect the gas outlet to the mass flow meter & reset the mass flow meter to 0 mLn
 - *For online gasphase measurements:* Connect the microGC in bypass configuration between reactor and MFM
- Open the connection to the gas bag

Electrolysis

- Start electrolysis by switching on the DC source(s) - simultaneously: start a timer
- Immediately after start and then in 10-minute intervals (longer intervals at longer experiment duration), record the measured values: gas flow (MFM), cell voltage (DCsource), pH, conductivity, temperature & observations

- *For online gasphase measurements:* Name the measurement "experiment number_ *time*", e.g. 25057_10 min and start the measurement (takes 7 min 30 s) every 10 minutes
- Last measurement shortly before the end of the electrolysis
- After the electrolysis time is over: Switch off the DC source & record the pH/conductivity values without current + gas flow
- If acidification is planned (standard, besides experiments with carbonate as supporting electrolyte):
 - Attach a syringe to the gas inlet with 3 M H₂SO₄
 - Carefully adjust the pH to pH = 2 (if no pH meter is available: use the same volume of H₂SO₄ as for the replicate with pH meter) -> be careful regarding the evolving gas volume and maximal flow of MFMs
 - Note the final gas amount
- Flush the dead volume with nitrogen via the nitrogen line connection: open the gas inlet and carefully open the nitrogen (flow must not exceed 50 or 20 mL N₂ per minute, depending on the flow meter -> it is best to open another gas inlet for easier flow adjustment)
- Flush until approximately 150 mL have been detected and note the final gas value
- Close the gas bag
- Switch off the pump & stirring plate, wait till the phases separate and put the tube below the organic phase
- Flush the system for 10 min with the reaction solutions (turn on the pump on same reaction conditions)
- Empty the flow reactor by sucking air through the pump
- Rinse the flow reactor with ~ 5 mL cyclohexane by inserting it via the inlet tube (to accomplish this, the inlet tube is disconnected from the reactor, a three way valve is attached, subsequently 5 mL cyclohexane is injected using a syringe, the cyclohexane then flows into the flow reactor via the pump. As soon the flow reactor is filled with cyclohexane, the system is closed to retain the cyclohexane within the flow reactor. The flow reactor is then shaken, after that the three way valve is opened, allowing the cyclohexane to flow into the anolyte reservoir via pump. (This same procedure is used for all rinsing steps)
- Rinse the reactor with ~ 5 mL cyclohexane by inserting it via the inlet tube
- Rinse the reactor with ~ 5 mL H₂O by inserting it via the inlet tube
- Remove the cables, hoses, tubes and sensors from the flow reactor and reservoirs
- Flush the flow reactor and the tubings with acetone and distilled Water, to remove residual water, the reactor is flow reactor with nitrogen.
- Ready for next reaction (with new tubings) or if no further reactions are planned, the flow reactor can be disassembled (disassemble the flow reactor and rinse the electrodes with acetone and distilled water)

Sample preparation & analysis

Phase separation and density determination

- Pour the reaction solution into separation funnel
- Wait for phase separation without shaking the funnel
- After phase separation, the aqueous phase is collected in the anolyte reservoir used, the organic phase is collected in a weighed test tube
- The funnel is rinsed with 5 mL of cyclohexane, which is subsequently transferred to the aqueous phase in the anolyte reservoir

- The analyte reservoir is sealed and shaken, subsequently, the solution is poured back into the funnel, and the funnel is then shaken
- Wait 10 min for phase separation, the aqueous phase is collected in a weighted bottle, and the organic phase in the test tube
- Weigh total organic and aqueous phases in flasks
- Take 1 mL of aqueous and 1 mL of organic phase after reaction (into weighed empty GC vial)
- Weigh the 1 mL samples to determine density (also t0)

Sample preparation for organic phase (GC)

- Label three GC vials with "experiment number - dilution factor" (dilution factors: 10, 100, 1000) or print more detailed labels
- Fill 200 μ L of the organic phase sample in test tube and add 1.800 μ L cyclohexane (dilution factor: 10)
- Repeat this step with the diluted solution (df10 and df100) 2x that you get diluted solutions with df 100 and df 1000
- Fill 100 μ L internal Cyclohexane standard FAME (should be out of the fridge for 1 h, close after usage, and use the same for the whole set) and add 900 μ L of the single solutions (3x diluted / df10 / df100 / df 1000) in labeled GC vial
- Close GC vial with cap and test for tightness
- First, the samples are analyzed using an HP-1 column in a GC-MS to monitor the alkanes formed. Subsequently, the samples are placed in another GC-MS to detect the resulting alcohols using a DB-FATWAX column

Sample preparation for aqueous phase (t0 & t_end) (HPLC)

- t0
 - Label two HPLC vials with "experiment number - dilution factor" (dilution factors: 100/1000 (optimized for single compound solutions $\leq$ n-C6, to be revised for mixtures / longer acids))
 - Fill 10 μ L of the t0 sample in first vial and add 990 μ L 5 mM H2SO4 (dilution factor: 100), close the vial and vortex
 - Fill 1 μ L of the t0 sample in the second vial and add 999 μ L 5 mM H2SO4, close the vial and vortex -> df1000
- t_end
 - Label two HPLC vials with "experiment number - dilution factor" (dilution factors: 2/20 (optimized for single compound solutions $\leq$ n-C6, to be revised for mixtures / longer acids))
 - Fill 500 μ L of the t_end sample in first vial and add 500 μ L 5 mM H2SO4 (dilution factor: 2), close the vial and vortex
 - Fill 50 μ L of the t_end sample in first vial and add 950 μ L 5 mM H2SO4 (dilution factor: 20), close the vial and vortex

Per experiment:

- 3 GC samples (3x Cyclohexane),
- 4x HPLC samples (2x aqueous - t0, 2x aqueous - t_end)

Offline gasphase measurements

- The gases formed are measured by measuring the gases collected in the gas bag three times on a microGC (using the last measurement)
- BakeOut the microGC for 20 min at the beginning of the day

- Load latest calibrated Method: "MicroGC_standard_method_calibrated_0726" (or latest version)
- Attach gas bag to microGC
- Add tags like "Kolbe", "MP (or your initials)" to the measurements
- Name the measurement "experiment number_rep1-3", e.g. 25057_rep1
- Start the measurement (takes 7 min 30 s)
- Three measurements per gas bag
- Export the data as CSV-File

Membrane handling

- Pre-Treatment / Conditioning: rinse the membrane for 24 hours in 0.5 M NaCl solution build in in the counter chamber
- Storage for up to several weeks in 1 wt% NaCl solution (10 g / L)

Before reaction:

- membrane is rinsed with Milli-Q water to remove residual NaCl before being installed in the cell

After reaction:

- after the experiment(s), the membrane is threated with acetone and milliQ water and subsequently placed into the NaCl solution

Attachments

No file attachments

*This procedure was originally created by **Max Pohl***